

Introduction

I am a geophysicist who studies dynamic surface processes across the continuum of Earth and planetary environments using multidisciplinary analyses. I combine ground-based, airborne, and orbital remote sensing of rock glaciers and other indicators of past and present alpine cryosphere activity. My work has implications for characterizing water resources, landscape evolution, climate history, ecosystem health, and the mitigation of natural hazards.

As a trained planetary geophysicist specialized in rock glaciers on Earth and Mars, I have combined high-precision near-surface geophysical methods—particularly ground-penetrating radar (GPR)—with optical and microwave remote sensing data to investigate shallow subsurface structure and link surface processes with climatic phenomena. Permafrost comprises more than 20% of the land area of the Northern Hemisphere, and major river basins across Earth contain alpine permafrost that contributes to the streamflow of their tributaries. Furthermore, rock glacier velocity and active layer thickness are defined by the Global Climate Observation System as essential parameters indicative of permafrost change due to climate-related processes, but there are currently no standard ideal methods for characterizing these variables at regional to local scales. My research has linked measurements of internal structure and surface deformation rates to demonstrate the interplay between rock glacier velocity, active layer thickness, surface morphology, englacial stratigraphy, and Holocene climate fluctuations. Not only has this work advanced the field of alpine permafrost science on Earth, but I am also a Co-Investigator on a NASA Mars Data Analysis Program grant that was selected to apply the morphological framework I developed to the interpretation of radar sounding data and orbital imagery targeting subsurface ice deposits in the Martian midlatitudes. This project connects surface morphology with near-surface structure using optical and radar remote sensing, which will advance our understanding of paleoclimate, current surface dynamics, and habitability across the solar system.

My postdoctoral work focuses on the application of interferometric synthetic aperture radar (InSAR) to generate time series of rock glacier velocity at high temporal resolution over a wide geographic area. As a member of the NASA Surface Topography and Vegetation (STV) Science Team, I have developed a technique that fuses measurements of surface deformation derived from high-resolution optical photogrammetry and InSAR. This method leverages the advantages of each discipline to generate models of rock glacier velocity and quantify the joint uncertainty that is most consistent with the combined datasets. My work has resulted in the lead-authorship of three peer-reviewed publications and co-authorship of one peer-reviewed publication; I also have two lead-author manuscripts under review, two co-author manuscripts under review, and multiple manuscripts in preparation, including those authored by students that I mentor using recently acquired data from a field campaign that I led in the San Juan Mountains of Colorado.

I envision developing a permafrost and alpine surface processes research group with a mission that efficiently combines all available measurements in sensitive permafrost environments to characterize the processes that will play a role in ongoing societal challenges. This research group will operate using a multidisciplinary approach that broadly combines the three widely applicable techniques described in the following sections: in situ near-surface geophysics, remote sensing data analysis, and process-based modeling of permafrost dynamics.

1. In situ near-surface geophysics advances rock glacier science

While remote sensing data is extremely valuable in its ability to provide repeated measurements over a large spatial scale, it is inherently limited in measurement precision both in terms of spatial resolution and the ability to detect shallow subsurface structure. As a result, in-situ measurements increase the value of remote sensing missions by providing direct “tie points” that serve as references for extrapolating mission data over large spatial areas. I have extensive experience planning and executing campaigns to safely and efficiently collect surface-based geophysical data in the cold mountain environments where alpine permafrost is located. My primary method for characterizing subsurface rock glacier structure is GPR due to its ability to image cryostratigraphic contacts and measure the distribution of ice and lithics with a precision that cannot be achieved by remote sensing alone.

In addition to surface-coupled GPR, my team members and I have operated low-altitude drone-based GPR systems to measure near-surface structure, employed active and passive seismic methods to measure the mechanical properties of the surface debris layer, installed automated weather stations to log meteorological data, acquired high-precision positioning information for morphological features (such as melt streams, ice exposures, and marked boulders), performed terrestrial laser scans to correlate microtopography with subsurface and performed transient electromagnetic soundings to model the subsurface resistivity structure. My deep understanding of geophysical field methods allows me to discern the advantages and limitations of various techniques in a range of periglacial settings. As a faculty member, I would plan annual monitoring campaigns to alpine permafrost sites in the western US to contribute to the growing inventory of rock glaciers with available near-surface geophysical data. These annual fieldwork expeditions would ensure that “ground-truth” calibrations supplement remote sensing and modeling analyses, and the geophysical techniques that I employ are broadly transferable to a host of other environments and surface processes.

2. Multi-sensor fusion optimizes the strengths of remote sensing techniques

Ground-based near-surface geophysical data in alpine permafrost is valuable because of the precision and detail that it returns, but the costs of in-situ data collection include variable accessibility to different sites of interest and the expenses required to make measurements over short intervals. Remote sensing—in the form of mid-altitude drone flights, piloted airborne missions, and orbital repeat-pass observations—affords the ability to expand survey areas and reduce time intervals between measurements, enabling a broader and more complete characterization of the processes that govern alpine permafrost surface dynamics. As an example, my work integrating drone-based and piloted photogrammetry datasets with satellite-based optical imagery produced detailed surface velocity fields, elevation change models, and strain rate maps for four rock glaciers. The ground-based positioning calibration allows for the characterization and minimization of measurement uncertainty and bias. Together with GPR measurements of surface debris thickness, englacial stratigraphy, and basal geometry, I documented the complex relationship between rock glacier morphology and its dynamic history.

At Galena Creek, Wyoming, I showed that flow-transverse ridge morphology in alpine permafrost can be generated through multiple mechanisms identifiable through near-surface geophysics, including compressional stresses and oscillations in climate that lead to interbedded ice and debris layers. Moreover, the high-resolution strain maps and elevation change models revealed that localized extension reduces overburden thickness, thus reducing its thermal insulation capacity and leading to faster subsidence due to increased subsurface melt in

concentrated areas. This work will inform ongoing characterizations of active alpine permafrost, the Quaternary history of the mountain cryosphere, and the degree to which these environments are changing in the 21st century. My research will play a role in discussions about water resources and the associated societal impacts.

Recently, I have focused on integrating orbital InSAR with the aforementioned methods for a more complete characterization of rock glacier populations and their surface motion. This method, which is widely relevant for disciplines ranging from seismology to soil science, affords rapidly recurring measurement intervals over large spatial extents compared to typical field-based observations (e.g., 12-day repeats for Sentinel-1 data). As a result, InSAR is ideal for measuring seasonal deformation and monitoring regional variability in alpine permafrost movement. While this method is highly precise along the satellite's line of sight to the surface, any motion perpendicular to the line of sight is not detectable; further, InSAR is limited by its relatively coarse spatial resolution compared to photogrammetric methods.

To address these challenges, I developed an algorithm for combining Sentinel-1 line-of-sight velocity results from multiple orbital geometries with the north component of independent surface velocity measurements to constrain the magnitude and spatial distribution of rock glacier deformation with greater fidelity than previous individual measurements. This algorithm is a direct example of the efficacy of the multi-tiered approach to rock glacier research: high spatial resolution field data and high temporal frequency orbital data complement one another to generate the most comprehensive understanding and robust uncertainty quantification of global permafrost dynamics. I am the Science PI on a NASA Decadal Survey Incubation proposal, which was selected to continue this work, and our lab has recently conducted fieldwork in Colorado to supplement our ongoing InSAR data analysis. InSAR will continue to be an important tool across the geosciences and planetary sciences, particularly with the recent successful launch of the NISAR mission. I am eager to be involved with the analysis of alpine permafrost targets as NISAR data becomes publicly available, and I plan to seek ongoing funding opportunities that will support this objective.

3. Modeling permafrost's role in changing Earth and planetary environments

Synthesizing multiple datasets ranging from subsurface to orbital is simultaneously an opportunity and a challenge. Measurement and processing requirements vary widely for different techniques, and each is associated with distinct physical parameters that contribute to the understanding of the rock glacier system. Interpretations of geophysical data and computational models designed to test hypotheses of rock glacier evolution must meet the conditions and error envelopes associated with the available datasets. These combined interpretations will provide tighter constraints on the landscape evolution of alpine permafrost. For instance, coupling present-day velocity fields with GPR-based landform thickness measurements provides limits on the rheological properties governing rock glacier deformation, which is consequential for estimating landform age and mass wasting rates. Combining meteorological data with debris thickness measurements, topographic models, and high-resolution time series derived from InSAR reveals how rock glaciers variably deform over seasonal timescales, with contributions from temperature, precipitation, and slope aspect.

I am interested in continuing these inverse modeling projects to study the global variability of alpine permafrost evolution and to continue to adapt algorithms to obtain measurements of essential climate variables. I have developed a method for monitoring rock glacier velocity, and I have begun to expand this method to study sites in North America, South

America, and Europe. I am also interested in modeling the remotely sensed active layer thicknesses using GPR and thermal conduction measurements as in situ calibration points. These processing workflows and modeling frameworks are computationally intensive while also requiring geological reasoning and decision-making; therefore, I intend to recruit mentees from a variety of backgrounds to hone their expertise towards the most impactful alpine permafrost projects. For example, a mathematician could adapt a finite elements model to simulate rock glacier dynamics, a computer scientist could build software that streamlines the InSAR processing algorithm to update rock glacier inventories, and a geospatial analyst could work on mapping rock glaciers as terrestrial analogs and compiling a comprehensive database of morphologies indicative of past flow and climate variation. All these projects would benefit from the existing Geophysical Computational Facilities and the Near Surface Geophysics Instrumentation Center at the University of Wyoming, and the scope of this work is relevant for acquiring funding from agencies such as the NSF, NASA, and the Army Research Office. I am confident in my ability to lead this research by example, and I am excited by the opportunity to build a group of energetic early-career geoscientists with a diverse range of skills and interests. This interdisciplinary research group would directly contribute to the scientific community and, by extension, increase the available data with which our global communities can make informed decisions.

Complementarity with the research community

My overarching research goal strives to integrate all available measurement methods to characterize the dynamic processes shaping the frozen regions of the Earth's surface. These processes contribute to the resources, habitats, and hazards that our society interacts with, and I am determined to be a contributor to the observations and communications that empower our communities to make decisions amidst these changing surface processes. This interdisciplinary and collaborative approach is aligned with the mission of the Department of Geology and Geophysics, and the methods that I employ are widely adaptable to research focused on wildfire, snow science, climate and water resources, agriculture and land use, and natural hazard assessment. My research focused on the cryosphere of Earth and Mars would complement multiple existing research groups, including Geohydrology, Geophysics & Seismology, and Surface Processes & Cold Regions. My research interests, combined with my background in geophysical field methods and radar remote sensing, would give me a unique perspective as a faculty member contributing to the Department of Geology and Geophysics. I would be eager to build collaborations within the department and network further with others amongst the diverse range of expertise found throughout the University of Wyoming. Recent advances in technology have led to a proliferation of active instrumentation and increased data availability. The community must leverage this abundance of overlapping data sets to generate the most detailed possible understanding of alpine permafrost surface dynamics and its role in broader environmental systems; I am uniquely positioned to lead this effort over the next several decades.